ML CAPSTONE · MEDICAL IMAGING

Kidney Stone Second Reader

Helping the doctor catch the stone they would miss, and measuring honestly how often it does.

[Your name] · [Date]

THE PROJECT IN ONE SLIDE

A second reader that points at kidney stones on CT

Input

One coronal non-contrast CT slice of the kidneys.

→

Output

Every candidate stone boxed and numbered. Red is confident; orange means look closer.

→

On new patients

Finds 93 of every 100 stones, with about one wrong box every 7 slices.

THE PAIN POINT

The first stone is easy to see. The next one gets missed.

10 px

median stone size on a 391 × 320 slice

52%

of slices hold two or more stones

13

stones in the most crowded slice

Source: audit of 778 coronal non-contrast CT slices, 1,699 labeled stones

THE USER

The doctor reading the scan

Who

A radiologist or ER doctor reading a non-contrast CT for suspected renal colic, often alone on a night shift.

What they do

1. Open a coronal slice.
2. See every candidate stone boxed, numbered and scored.
3. Accept or reject each box, then count the stones.

It proposes. The doctor decides.

Red: right 9 times in 10

Orange: look closer

FRAMING THE PROBLEM

Four choices, all from the pain point

We chose	Because
Detection, not yes/no classification	The pain is the missed second stone: the doctor needs *where* and *how many*.
Score **per stone**, not per image	An image with 4 stones and 1 missed is still a miss for the patient.
Split **by patient**, not by image	The real question is how it does on a patient it has never seen.
High sensitivity, few false alarms	A missed stone costs more than a box to dismiss, but too many boxes and doctors stop looking.

Constraints: one day, one person, a free Colab T4 GPU, 778 labeled slices

THE PIPELINE

From CT slice to second reading in six steps

1 · Acquisition

Coronal non-contrast CT, cropped to the kidneys, 391 × 320 px. Stones boxed by an annotator.

2 · Curation

Merge augmented copies. Recover 256 scan sessions from the DICOM acquisition time.

3 · Patient-level split

70 / 15 / 15 by scan session. No patient appears in two sets.

4 · Detection model

YOLOv8s pretrained on COCO, 960 px input, flip, rotation and mosaic augmentation.

5 · Post-processing

Merge overlapping boxes. Two thresholds learned on validation decide red, orange or hidden.

6 · Evaluation and delivery

Per-stone metrics and FROC on unseen patients, then a Gradio second-reader app.

THE DATA

What the dataset holds, and what it hid

778
unique slices in 1,300 files

256
scan sessions, our stand-in for patients

1,699
labeled stones, one class

1
slice without a stone

The published split mixes patients

93 of 256 scan sessions have slices in both train and test, and 68 of 123 published test slices come from patients seen in training. So we split by patient.

Roboflow “TEZ_ROI_AUG” v3, CC BY 4.0 · Philips CT, Aug 2019 to Dec 2020

TRAINING AND TUNING

Tuned for the doctor’s workload, not the leaderboard

Test set, unseen patients	v1	v2 (final)
Input size	640 px	960 px
Box merging (NMS IoU)	0.7	0.3
Show threshold	0.020	0.089
Stone sensitivity	94.8%	**93.1%**
False alarms per slice	0.73	**0.13**
mAP50	0.732	0.632

v1 problem: stacks of 5 to 10 boxes on one stone, and a threshold so low (0.02) that most boxes were orange.

v2 fix: merge overlapping boxes and learn the red cutoff from validation. False alarms fell about 5 times.

The cost: mAP50 fell 10 points. The 960 px input did not raise validation mAP (0.737 to 0.713).

v2: 60 epochs, 29 min on a T4, training loss converged · v1 scored with a stricter centre-only hit rule, so its row is approximate

HOW WE MEASURE IT

Metrics a doctor would ask for

Stone sensitivity

Of all real stones, the share that got a box.

False alarms per slice

Boxes pointing at nothing that the doctor has to dismiss.

Precision

Of all boxes shown, the share that finds a stone.

All stones found

Share of slices where not one stone was missed, so the count is right.

FROC curve

Sensitivity against false alarms at every threshold, the radiology standard.

mAP50

The standard ML detection score, so results compare with other projects.

A hit = box centre inside the stone, or overlap IoU ≥ 0.1 · thresholds chosen on validation only · 95% intervals resample whole patients

RESULTS · 120 SLICES FROM 39 UNSEEN PATIENTS

93% of stones found, few false alarms

93.1%
stone sensitivity (95% CI 88 to 98%)

0.13
false alarms per slice

90.8%
of slices with every stone found

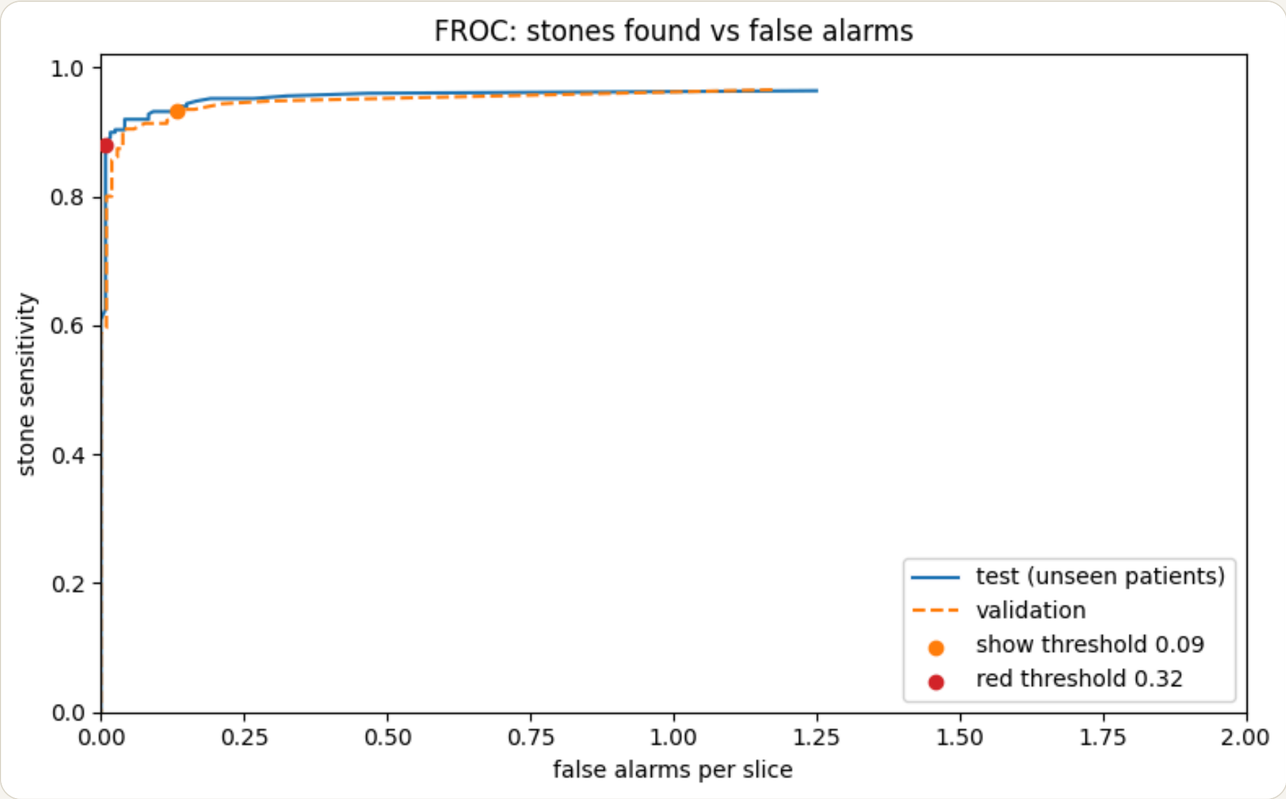
0.63
mAP50 (ML metric)

Boxes shown	Sensitivity	False alarms / slice	Precision	All found
Red + orange (default)	**93.1%**	**0.13**	**81%**	**90.8%**
Red only	87.9%	0.008	94%	82.5%

Test set: 248 stones · show threshold 0.089 and red threshold 0.321, both chosen on validation

EVALUATION

The trade-off, and why we trust it



Validation and test agree. The dashed and solid curves overlap, so thresholds chosen on validation carry over to new patients.

Red dot: red boxes alone find 88% of stones at almost no false alarms.

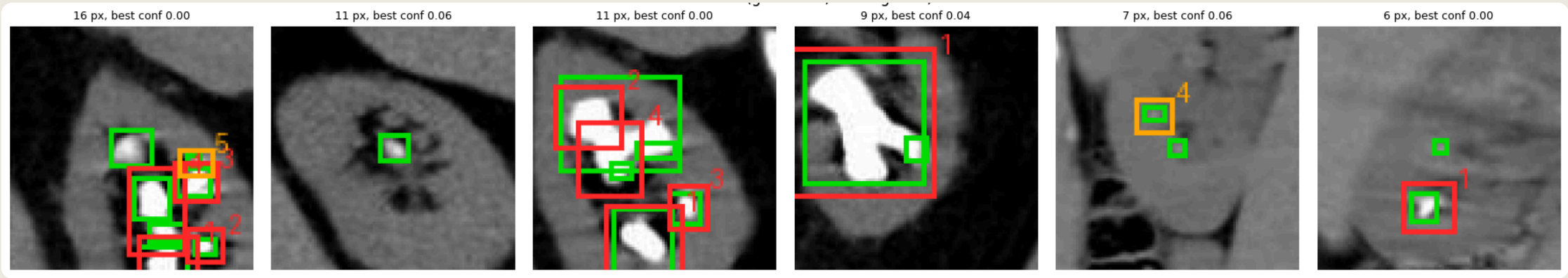
Orange dot: adding orange boxes reaches 93% at 0.13 false alarms per slice.

ERROR ANALYSIS

It misses small stones and crowded kidneys

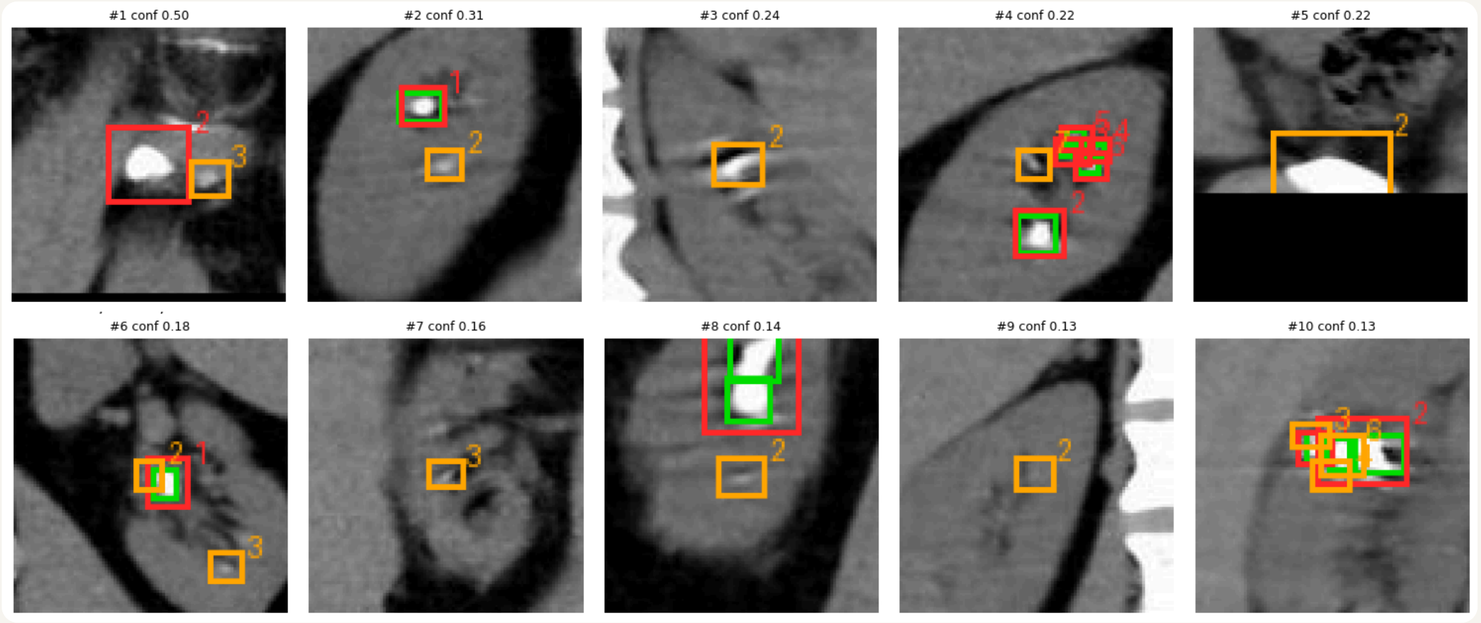
Stone size	Median	Found
Small	6 px	84%
Medium	10 px	96%
Large	19 px	99%

Stones in slice	Slices	All found
1 or 2	90	97%
3 to 4	20	90%
5 or more	10	40%



LABEL AUDIT

Are the false alarms really wrong?



2 look like real stones the labels missed (#1, #5)

2 are boxes just beside a labeled stone (#6, #10)

6 are faint spots or vessel-like streaks: mimics or noise

The 10 most confident false alarms on the test set · author’s visual read, not radiologist-verified

CLINICIAN OUTCOME

What it means at the reading desk

93 of 100 stones get a box

The 7 it misses are mostly small, the kind that usually pass on their own.

One wrong box every 7 slices

Few enough that the doctor keeps trusting it instead of switching it off.

Red boxes can be trusted

One false red box in 120 slices. Orange boxes ask for a second look and catch 5% more stones.

The second stone

Every stone found in 97% of slices with 1 or 2 stones, 90% with 3 to 4. Weak above 5.

Expected benefit: fewer missed stones and faster confirmation. Not yet proven with doctors: that needs a reader study.

BEFORE REAL USE

Eight things to fix before a real clinic

Specificity unknown

No stone-free slices, so false alarms on healthy patients are unmeasured.

One source

One center, one vendor, 2019 to 2020. Needs external validation.

2D, not 3D

Doctors scroll the whole volume; one stone spans several slices.

No size in mm

Pixel spacing was lost, so it cannot guide pass-or-treat decisions.

Kidney crop only

Ureteral stones, the usual cause of colic, are out of scope.

Noisy labels, mimics

One annotator; some stones unlabeled. Phleboliths can fool it.

Not tested with doctors

Needs a reader study: doctors reading with and without it.

Bias and regulation

Doctors may trust its misses. It is a medical device and needs approval.

LIVE DEMO

It proposes.
The doctor decides.

Upload a coronal slice. Get every candidate stone boxed and numbered, a stone count, and a confidence for each one.

Red: right 9 times in 10

Orange: look closer